

Function Vectors Are Two Populations: Writers and Cancellers in ICL Heads

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Abstract

Function-vector (FV) heads (Todd et al., 2024) are typically identified by the *magnitude* of their causal contribution to in-context rule tasks, under the implicit assumption that the top set is a homogeneous functional class. **This assumption fails.** Replacing magnitude-only ranking with a sign-preserving criterion (refined DLA + permutation FDR) and validating each candidate by path patching, we find the FV head population splits into two opposing sub-populations: *writers* push the rule-correct logit up; *cancellers* push it down. The four-condition canonical verdict holds across three model families and six Pythia scales (13/15 cells canonical), and a sign-shuffle null rejects population homogeneity. The structure is invisible to magnitude-only ranking: on the hierarchical-rule task, Todd’s top-20 captures 64% of cancellers but only 4% of writers; on the modular-rule task, 59% of writers but only 8% of cancellers. Six artefact accounts are ruled out across all 27 canceller (cell, head) pairs across the six main cells: induction overlap, sinks, generic importance, rank-1 copy-suppression, V-cascade, and rank-nearest non-FV controls. Zero-ablating cancellers produces a +0.13 to +0.29 nats logit gain in 6/6 main cells with a directionally consistent +2–7 pp accuracy effect.

1 Introduction

Function-vector (FV) heads (Todd et al., 2024; Hendel et al., 2023; Variengien and Winsor, 2023; Olson et al., 2022)¹ have become a central object of study in mechanistic interpretability: a sparse set of attention heads whose causal contribution to

¹Glossary. *FV head*: a function-vector head. *DLA* (direct logit attribution): a head’s signed contribution to the correct–incorrect logit difference (Geva et al., 2022). *OV*: the head’s write-direction in the residual stream; *QK*: the head’s attention-source pattern (Elhage et al., 2021). *Path patching* (Wang et al., 2023; Goldowsky-Dill et al., 2023) isolates a head’s *direct* effect by patching its activation while holding downstream paths fixed.

in-context-learning (ICL; Brown et al., 2020) rule tasks is large enough to support targeted steering, ablation, and circuit-level claims. Inventories built from these heads now ground a growing literature on what models “store” about in-context tasks. The standard pipeline ranks heads by the *magnitude* of a causal contribution metric and treats the top set as a homogeneous functional class. Implicit in this pipeline is a sign assumption: each high-magnitude head pushes the model *toward* the demonstrated rule. Under this assumption, magnitude is identity, and the head population can be summarised by a single number per head.

We show, across the Pythia ladder (410M–12B, six sizes) and two ICL rule tasks plus three spot-check models from different families (Qwen2.5- $\{1.5, 7\}$ B, GPT-2-medium), that this assumption fails empirically. The high-magnitude population splits cleanly into two opposite roles: *writers* (refined-DLA > 0 , push the rule-correct token’s logit up) and *cancellers* (refined-DLA < 0 , push it down). The split is large, reproducible across cells, and not explained by attention-sink, induction-overlap, copy-suppression, V-cascade, generic-importance, or rank-nearest non-FV head confounds. We lead with the population finding; the L11.H4 case study (§4.8) is one illustrative head.

Approach. For each (model, task) cell we identify the FV head set via refined DLA, validate causal effects with path patching, partition by sign of direct effect into $\mathcal{W}$ and $\mathcal{C}$, and run a four-condition canonical verdict ($W_{\text{shift}} < 0$, $C_{\text{shift}} > 0$, opposite signs, joint attenuated). The attention-sink confound is ruled out via per-source DLA bucket attribution, joined by five further population-level rule-outs (induction-overlap TOST, generic-importance ratio, rank-1 copy-suppression (McDougall et al., 2023), V-cascade (Wang et al., 2023), rank-nearest non-FV head). A complementary

single-head case study localises the dominant canceller L11.H4 in Pythia-410M with three orthogonal interventions (V-shuffle, OV singular spectrum, and V-composition with the dominant writer) and asks which standard mechanistic template it instantiates.

Related work. Sign-opposed attention heads have appeared in mechanistic interpretability as *single-task, hand-built circuits*: IOI’s negative name-movers (Wang et al., 2023) and GPT-2 L10.H7’s rank-1 copy-suppression (McDougall et al., 2023). The FV-head literature (Todd et al., 2024; Hendel et al., 2023; Variengien and Winsor, 2023; Subramani et al., 2022) operates one level up on the head *population* but ranks by causal magnitude, which collapses sign. **We lift the opposing-heads phenomenon to a structural property of the FV head population**, identified by sign-preserving DLA screening and validated by path patching, sign-shuffle null, and six rule-outs across 15 (model, task) cells; the L11.H4 case study extends copy-suppression to ICL but neither standard template fits its mechanism. Adjacent SAE-feature work (Bricken et al., 2023) is complementary. Tools: direct-logit attribution (Geva et al., 2022), path patching (Wang et al., 2023; Goldowsky-Dill et al., 2023), causal mediation (Vig et al., 2020; Meng et al., 2022); mechanism framing: OV/QK (Elhage et al., 2021); circuit catalogues (Hanna et al., 2023; Nanda et al., 2023; Conmy et al., 2023; Syed et al., 2024; Marks et al., 2025).

Contributions. We make four claims, each independently verifiable. **First, an empirical population finding:** the FV head set contains two opposing functional roles (refined-DLA writers push the rule-correct logit up, cancellers push it down), and the four-condition canonical verdict ($W_{\text{shift}} < 0$, $C_{\text{shift}} > 0$, opposite signs, joint attenuated) holds in 13/15 cells across the Pythia ladder (410M–12B) and three architectural spot checks (Qwen2.5- $\{1.5, 7\}$ B, GPT-2-medium); 2/15 remain partial. A sign-shuffle null ($n_{\text{perm}} = 10,000$) rejects population homogeneity at family-wise $\alpha = 0.05$ (Holm-Bonferroni) in 5/6 main cells. **Second, six population-level mechanism rule-outs** on $n = 27$ canceller (cell, head) pairs across the six main cells: induction-head overlap (TOST equivalence to chance, 6/6); BOS/format-prefix sinks ruled out (74% of cancellers are content-driven, not sink-driven); generic-importance ($5\times\text{--}151\times$ rule-specific

lesion ratio); rank-1 copy-suppression (McDougall et al., 2023) (0/27 above threshold); V-cascade (Wang et al., 2023) (ruled out in 4/6); a rank-nearest non-FV control (FV ablations mean $5.5\times(W)/5.2\times(C)$ larger than $|\text{DLA}|$ -matched random partitions). Writer and canceller mean OV directions are nearly orthogonal (perpendicular fraction mean 0.96, bootstrap 95% CI $[0.91, 0.99]$). **Third, a candidate single-head mechanism:** L11.H4 (Pythia-410M, present in both rule tasks) is a *distributed-bias content reader*: V-shuffle collapses its DLA by $\sim 82\%/ \sim 53\%$, $W_O W_V$ has top-1 Frobenius share 2.8%, and ablating its dominant writer L10.H9 leaves its DLA unchanged. The same head flips role from canceller to writer on antonym under cross-template transfer (canceller group $\Delta\ell = -1.56$ nats), the expected behaviour of a “suppress what was demonstrated” mechanism. We do not extrapolate this single-head template to the full canceller population. **Fourth, an immediate calibration-free intervention:** zero-ablating cancellers produces a logit gain of $+0.13$ to $+0.29$ nats with CI excluding 0 in 6/6 main cells, and a directionally consistent $+2\text{--}7$ pp accuracy effect (CI-confirmed on hier-410M) — not via the Todd-style aggregation route, which we show is the wrong operation for this purpose (§5).

2 Setup

Models. The main pipeline studies three Pythia checkpoints (Biderman et al., 2023) (410M, 1B, 1.4B), loaded in fp32 with `attn_implementation=eager`. Pythia provides a single architecture family across an order-of-magnitude scale range, isolating the scale axis from architectural confounds. §4.6 extends to the rest of the Pythia ladder (Pythia-2.8B/6.9B/12B; bf16 on the ≥ 6.9 B sizes), and adds three single-task spot checks in different model families: Qwen2.5-1.5B/7B (Llama-style self-attention) and GPT-2-medium (Conv1D-style attention) on modular only, covering non-NeoX attention architectures.

Tasks. Two ICL rule-following tasks share a 4-shot surface form $f_0, f_1, Y \dots f_0q, f_1q$, with $(f_0, f_1) \in \{0, \dots, 7\}^2$, labels $Y \in \{A, B\}$, and a hidden rule $z \in \{0, 1, 2, 3\}$ fixed within a prompt. We use $K = 4$ in-context examples; feature pairs (f_0, f_1) are sampled i.i.d. across demonstrations within a prompt ($\beta = 1$: no repetition across the K demos). *Hierarchical*: z selects one of $\{f_0 >$

Model	arch.	L	H/layer	d_{model}	d_h
Pythia-410M	GPT-NeoX	24	16	1024	64
Pythia-1B	GPT-NeoX	16	8	2048	256
Pythia-1.4B	GPT-NeoX	24	16	2048	128
Pythia-2.8B	GPT-NeoX	32	32	2560	80
Pythia-6.9B	GPT-NeoX	32	32	4096	128
Pythia-12B	GPT-NeoX	36	40	5120	128
Qwen2.5-1.5B	Llama-style	28	12	1536	128
Qwen2.5-7B	Llama-style	28	28	3584	128
GPT-2-medium	Conv1D	24	16	1024	64

Table 1: Models studied. The first three rows form the main mechanism pipeline (Pythia 410M / 1B / 1.4B); the remaining six are extension cells used for behavioural-verdict transfer (§4.6).

3, $f_1 > 3$, $f_0 + f_1 > 7$, $f_0 = f_1$ }. *Modular*: z selects modulus $m \in \{2, 3, 5, 7\}$ (coprime, to avoid rule-subset issues at $\{2, 3, 4, 5\}$); the label is $\mathbf{1}[(f_0 + f_1) \bmod m = 0]$. The two tasks share surface form but differ in rule mechanism, so a head population that survives both is recruited by the rule, not the template.

Notation. Let (L, H) index a head, with $h_{(L,H)}(x) \in \mathbb{R}^{d_h}$ its output at the final query-token position before $W_O^{(L)}$. W_U is the unembedding matrix; γ is the final-LayerNorm gain; $r(x)$ is the pre-LN residual. We aggregate correct/incorrect-label probabilities over leading-space variants (e.g. $\{ \text{'A'}, \text{'A'} \}$ for label A) and define the unembed contrast $u(x) = W_U[y_+] - W_U[y_-]$, the readout $\Delta\ell(x) = \log p(y_+ | x) - \log p(y_- | x)$, and the per-pair swing $\Delta\ell(x_c) - \Delta\ell(x_r)$ between correct- and rule-flipped prompts.

Three prompt sets are sampled per (model, task) cell, with seeds fixed in advance: 192 balanced *discovery* prompts (used for refined-DLA, seed 42), 200 paired *path-patching* prompts (seed 43), and 500 paired *evaluation* prompts (seed 44) used for group lesions, cross-task transfer, and specificity. Logits are aggregated over the leading-space label variants above. Full prompt examples and the format-prefix string are in App. A.2.

3 Methodology

The pipeline runs independently per (model, task) cell. *Discovery* identifies the candidate FV head population (§3.1–3.2). *Causal arbitration* tests the cancellation signature at the group level against three independent nulls (§3.3). *Mechanism* characterises the writer/canceller relationship via attention sources (§3.4). *Robustness* tests that W/C labels are intrinsic to heads, not artefacts of the dis-

covery data (§3.5). *Specificity* controls distinguish writers/cancellers from induction heads (§3.6). All thresholds and the 2×3 Pythia grid are fixed in advance of the primary analysis.

Statistical conventions: paired-prompt percentile bootstraps with $B = 10,000$, BH-FDR (Benjamini and Hochberg, 1995) at $q = 0.10$ for cross-cell aggregation; full per-test inventory in App. A.1.

3.1 Refined Direct Logit Attribution

We attribute $\Delta\ell$ to head (L, H) via projection of its post- W_O contribution onto the unembed difference using a frozen-LN linearisation (Geva et al., 2022; Wang et al., 2023):

$$\widehat{\text{DLA}}_{(L,H)}(x) = u(x)^\top \left(\gamma \odot \frac{c_{(L,H)}(x) - \overline{c_{(L,H)}}(x)}{\sigma(r(x)) + \varepsilon} \right), \quad (1)$$

where $c_{(L,H)} = W_O^{(L)}[:, Hd_h:(H+1)d_h] h_{(L,H)}$, $u(x) = W_U[y_+] - W_U[y_-]$ is the unembed difference of the correct vs. incorrect label tokens, $\overline{c_{(L,H)}}$ is the batch mean over the discovery prompts (the LN linearisation reference), $\sigma(r(x))$ is the standard deviation of the pre-LN residual at the last token, γ the LN gain, and $\varepsilon = 10^{-5}$ the LN numerical stabiliser. The additive identity $\sum \widehat{\text{DLA}} + (\text{MLP, embed, biases}) = \Delta\ell$ is audited end-to-end (rel. tol. 5%). Per-head signed means are computed on the discovery batch (seed 42, $N = 192$) with a 20-seed label-permutation null and BH-FDR at $q = 0.10$ over $L \geq 1$. The screened FV-candidate set is the union of heads passing refined-DLA at this gate and the top- K heads by mean refined-DLA magnitude, with $K = \text{clip}(2n_{\text{FDR}}, 8, 50)$ where n_{FDR} is the per-cell count of refined-DLA passes (the clip ensures both small and large head populations yield a tractable shortlist).

3.2 Path Patching with Ratio-of-Means

We form 200 paired prompts (x_c, x_r) sharing query (f_{0q}, f_{1q}) but differing in rule (Wang et al., 2023; Goldowsky-Dill et al., 2023). After filtering pairs with $|\Delta\ell^{\text{full}}| < 0.5$, the ratio-of-means

$$\text{direct}_{(L,H)} = \frac{\mathbb{E}[\Delta\ell_i^{\text{direct}}]}{\mathbb{E}[|\Delta\ell_i^{\text{full}}|]} \quad (2)$$

(and analogously *total*, *indirect*) is bootstrapped ($B = 10,000$) for 95% CIs and a two-sided p_{direct} . PP p -values are BH-corrected at $q = 0.10$. The $\pm 5\%$ gate is fixed in advance; sensitivity to gate width ($\pm 3\%$, $\pm 7\%$, $\pm 10\%$) is reported in App. A.3. We also impose the two-hop ordering $L_W < L_C$ for any (W, C) pair entered into edge-level analysis.

Operational definition of writers and cancellers. $\mathcal{F}$ is the sign-preserving FV candidate set from refined-DLA screening (§3.1): unlike Todd et al. (2024)’s magnitude-only top- K , our criterion admits heads with both signs of direct effect, so it can in principle surface both writer- and canceller-type populations. A head $(L, H) \in \mathcal{F}$ is a *writer* if its direct PP effect is $\geq +5\%$ of the swing, a *canceller* if $\leq -5\%$, and *weak* otherwise; only writers $\mathcal{W}$ and cancellers $\mathcal{C}$ are entered into the group lesion (weak heads are listed in $|\mathcal{F}|$ for context but excluded from ablations). Per-cell counts appear in Table 3.

Asymmetric overlap with Todd’s magnitude-only set. Todd et al. (2024)’s mean-ablation top-20 captures only one of the two populations per task family (Table 2; full 6-cell breakdown in App. C.2). Magnitude-only ranking systematically surfaces whichever sub-population locally dominates and obscures the other; only sign-aware screening recovers both.

Table 2: **Todd top-20 overlap, pooled across 6 main cells.** Magnitude-only ranking surfaces opposite populations on the two task families.

task	W in Todd-20	C in Todd-20
hierarchical	1/27 (4%)	9/14 (64%)
modular	16/27 (59%)	1/13 (8%)

Writers and cancellers both interact with demonstration-label information but their attention distributions differ structurally: writers concentrate more mass on demonstration-label tokens, while cancellers shift mass toward format-prefix tokens (§4.2). The dominant negative DLA still comes from content rather than sinks (§4.3: 74% of cancellers content-driven by per-source DLA). Because the two groups write along different residual-stream directions, joint ablation produces an attenuated shift rather than the linear sum two independent same-direction heads would give.

3.3 Group Canceller Lesion: Headline Causal Test

For each ablation strategy $s \in \{\text{zero, mean}\}$ on $n_{\text{eval}} = 500$ paired prompts we ablate (a) writers $\mathcal{W}$, (b) cancellers $\mathcal{C}$, and (c) their union, bootstrapping 95% CI per group ($B = 10,000$). The canonical cancellation-circuit verdict requires: (i) $|\Delta\ell_{\mathcal{W}}| \geq 0.10$ with CI excluding 0; (ii)

$|\Delta\ell_{\mathcal{C}}| \geq 0.10$ with CI excluding 0; (iii) opposite signs; (iv) $|\Delta\ell_{\text{both}}| < |\Delta\ell_{\mathcal{W}}| + |\Delta\ell_{\mathcal{C}}|$. Sub-canonical patterns—writer-dominant, canceller-dominant, opposite-signs-without-attenuation, and co-writer—are recorded so non-canonical outcomes remain interpretable.

Two nulls. Conditions (i)–(iii) are mechanically forced by sign-partitioning; only joint attenuation (iv) is diagnostic. We test it against an FV-set sign-shuffle null and a |DLA|-matched head-randomised control (App. E.1: FV ablations mean $5.5\times(\mathcal{W}) / 5.2\times(\mathcal{C})$ larger and attenuation gap $2.5\times$ larger than |DLA|-matched random; the structural attenuation itself appears in matched random sign-partitions by construction, so the FV-specific evidence is magnitude). For $n_{\text{perm}} = 10,000$ draws we randomly partition $\mathcal{W} \cup \mathcal{C}$ into two subsets matching $|\mathcal{W}|, |\mathcal{C}|$, run zero-strategy ablation on each, and record the contrast $\Delta\ell^{\text{perm}\mathcal{W}} - \Delta\ell^{\text{perm}\mathcal{C}}$. We report two p -values per cell: the headline empirical p , and a Gaussian back-stop that guards against the 10^{-4} permutation-floor saturating the tail. The empirical two-sided p is the fraction of permutations with $|\text{contrast}| \geq$ the observed value, floored at $1/n_{\text{perm}} = 10^{-4}$; the Gaussian fit $p_{\text{Gauss}} = 2\Phi(-|z|)$ with $z = (\text{obs} - \mu)/\sigma$ uses the same draws but bypasses the floor (Table 6 columns 6–7). The headline FDR control uses the empirical p .

3.4 QK Source Decomposition

Each head’s last-position attention mass is partitioned into five buckets (BOS, format prefix, demonstration input, demonstration label, query input) and averaged over $n_{\text{prompts}} = 32$ eval prompts to produce a per-head bucket distribution. We report the per-bucket mean attention mass within $\mathcal{W}$ and $\mathcal{C}$ and their per-bucket difference (canceller minus writer) per cell. We ask whether the canceller-minus-writer sign is consistent across cells on the demonstration-label and format-prefix buckets.

3.5 Robustness and specificity setup

This subsection defines the auxiliary tests reported in §4.4: split-half stability, cross-task transfer, the within-layer OV null, and the TOST equivalence margin used by the induction-overlap rule-out.

Split-half. $n_{\text{split}} = 5$ random 50/50 splits of the eval prompt superset are drawn; on each, the W/C set defined on the full PP run is held fixed and the four-condition verdict is computed on the held-

out half B . A cell passes iff $\geq \lceil n/2 \rceil$ splits reach canonical on B .

Cross-task transfer. For each ordered task pair (A, B) the W/C labels from A 's PP are applied to B 's eval prompts. We use a *relaxed* variant that requires conditions (i)–(iii) (signed magnitude ≥ 0.10 nat, opposite signs) but drops condition (iv) (the joint-attenuation point inequality $|\Delta\ell_{\text{both}}| < |\Delta\ell_{\mathcal{W}}| + |\Delta\ell_{\mathcal{C}}|$). Cross-task transfer adds variance; we test sign persistence rather than the strict attenuation inequality. Cells that pass (i)–(iii) but fail the strict (iv) gate are reported as the writer-dominant distributed-cancellation pattern and counted separately from canonical (Table 8).

Within-layer OV null. For each (W, C) pair we sample ≥ 100 random co-layer head pairs (L_W, L_C) with neither head in $\mathcal{W} \cup \mathcal{C}$, compute the OV-projection cosine between them, and define the one-tailed p -value as the fraction of null cosines $\leq$ the observed cosine (more anti-aligned). A pair is called significantly anti-aligned if $p \leq 0.05$ (floor $1/100 = 0.01$). A cell passes the OV gate (Table 8) if at least 30% of its (W, C) pairs are significantly anti-aligned.

TOST equivalence margin. For the induction-overlap test, the equivalence margin is fixed at one hypergeometric standard deviation: $\Delta = \sqrt{\text{Var}_{\text{hg}}(K, n, N)}$ where K, n, N are the top-10 size, $|\mathcal{W} \cup \mathcal{C}|$, and total head population per cell. A cell is classified as equivalent to chance iff both one-sided p -values reject at $\alpha = 0.05$; as having no overlap iff the observed intersection is 0 (in which case TOST is degenerate and we report the empty-intersection verdict directly); and as inconclusive otherwise.

3.6 Specificity Controls

Cancellers vs. general importance. Each canceller is zero-ablated on rule-task and random-token sequences; we bootstrap 95% CI on $\mathbb{E}[\Delta\text{NLL}^{\text{rule}}]/\mathbb{E}[\Delta\text{NLL}^{\text{rand}}]$ ($B = 5000$). *Per-head verdict:* rule-specific iff ratio ≥ 5 ; generally important iff < 1.5 . *Induction-head overlap.* The top-10 heads by prefix-matching score (Olsson et al., 2022) are intersected with $\mathcal{W} \cup \mathcal{C}$; the per-cell intersection size is compared to the hypergeometric expectation under random allocation.

4 Results

The FV head set has size $|\mathcal{F}| \in [12, 23]$ across the 6 main cells. Conditions (i)–(iii) of the canonical verdict are mechanically forced by sign-partitioning, so four load-bearing pieces of evidence remain: (1) the canceller subgroup's causal-effect magnitude is mean $5.2\times$ larger than rank-nearest non-FV controls; (2) joint attenuation lies above a sign-shuffle null in 5/6 cells at $\text{BH-}q \leq 0.014$; (3) writers and cancellers are structurally separated (QK source, OV geometry, per-source DLA); (4) the overlap with Todd's magnitude-only top- K is task-asymmetric (§3.2).

4.1 Cancellers suppress the correct-label logit

Ablating the canceller subgroup shifts the readout $+0.13$ to $+0.29$ nats *toward* the correct label in 6/6 cells; ablating writers shifts -0.25 to -0.99 nats *away* from it (Table 3; the 6 main cells in Figure 1). The four-condition canonical verdict (§3.3) holds in 11/12 outcomes across 6 cells $\times$ 2 ablation strategies (zero, mean); the single miss is (mod-1.4B, mean), where the canceller magnitude ($+0.09$) sits just below the 0.10 threshold. Table 8 reports the per-cell verdicts under all six independent tests this paper relies on, so partial failures (cross-task hier $\rightarrow$ mod-1B, mean-strategy mod-1.4B, OV gate hier-1B + mod-1.4B, Fisher edge enrichment 1/6) are surfaced rather than buried.

Magnitudes and budgets. Joint ablation attenuates the writer effect by mean 48% (range 21–100%). Writers outnumber cancellers 2:1 with $\sim 1.8\times$ larger per-head magnitudes; per-head canceller shift is uniform (CV 0.19 vs. 0.56). Per-cell budgets in App. B.3.

Geometric independence. Writer and canceller mean OV directions are nearly orthogonal in the residual stream: cancellers' projection along the writer direction has perpendicular fraction 0.84–1.00 (mean 0.96; bootstrap 95% CI [0.91, 0.99]; App. D.4). Cancellers therefore write to a near-orthogonal subspace while still exerting a negative direct effect on u — a geometric decoupling that lets the two populations co-exist at the same readout without circuit overlap, and explains why joint ablation is attenuated rather than fully additive.

The verdict survives a sign-shuffle null. Because the W/C sign definition mechanically forces sign-balance, the appropriate null is 10,000 random sign assignments *within* $\mathcal{W} \cup \mathcal{C}$ (zero-strategy

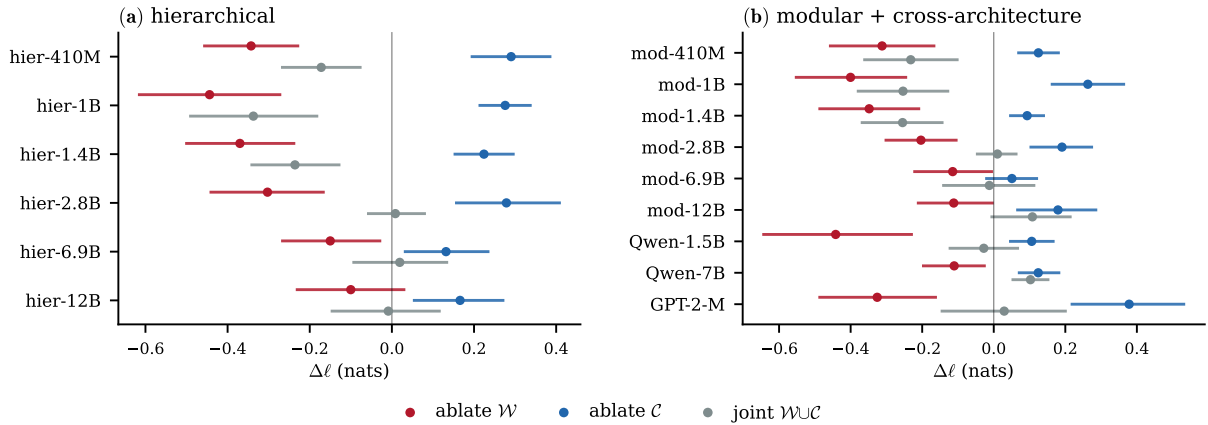


Figure 1: **Function vectors are two populations.** Mean-ablation logit shifts (BCa 95% CI) for writers, cancellers, and their joint, per cell. Writer ablation shifts the readout away from the correct label, canceller ablation toward it, and joint ablation attenuates. The four-condition canonical verdict holds in 13/15 cells. (a) Hierarchical (6 Pythia cells, 410M–12B); (b) Modular + cross-architecture (9 cells: Pythia mod 410M–12B, Qwen2.5-{1.5, 7}B, GPT-2-medium).

ablation). The observed $|W - C|$ contrast exceeds the null 95th-percentile in 5/6 cells, with empirical $\text{BH-}q \leq 0.014$ on those five; mod-1.4B is the single boundary case at $p_{\text{emp}} = 0.104$ ($p_{\text{Gauss}} = 0.088$, Table 6). Discussed in [Limitations](#).

A per-cell verdict matrix summarises six independent tests (zero 4c 6/6, sign-shuffle 5/6, split-half 6/6, cross-task 5/6); full breakdown in App. B.7. Per-cell magnitudes are reported in Table 3 (below, following §4.4).

4.2 Writers and cancellers read different prompt positions

QK-source decomposition rules out the hypothesis that cancellers are sign-flipped writer copies. Writers and cancellers read structurally different prompt positions, with the same direction of difference in 6/6 cells on the two load-bearing buckets (Figure 2; per-cell numerics in App. C.3, Table 11). Writers concentrate 34–52% of attention mass on demonstration labels and 28–44% on BOS; cancellers shift 0.09–0.17 of mass off the label position and gain 0.08–0.18 on format-prefix tokens (15–29% of canceller mass vs. 4–17% for writers). Cancellers are not the same heads with a sign flip.

4.3 Mechanism is content, not sinks

The format-prefix pattern raises a sink confound: cancellers might be format sinks whose effect is an artefact of where attention lands rather than what they read. Per-source DLA bucket attribution rules this out. We split each canceller’s direct effect by source bucket and label it *content* when its negative DLA is dominated by demonstration-content buck-

ets, *sink* otherwise. 20 of 27 (74%) cancellers are content-driven, and they carry the headline magnitude (−0.10 to −0.23 nats); the 7 sink heads cluster near zero (−0.02 to −0.09 nats) and reappear cross-task at fixed scale, consistent with format readers rather than rule-specific suppressors. Per-cell numerics in App. C.4, Table 12; an independent V-shuffle protocol on the strongest canceller per cell confirms content-driving in 6/6. L11.H4 is fully content-driven (~100% demonstration-label sourced in both 410M cells), foreshadowing the single-head mechanism analysed in §4.8.

4.4 Robustness and specificity

Split-half ($5 \times 50/50$ prompt splits): verdict stable 5/5 seeds in 6/6 cells. *Cross-task transfer* (W/C labels from src on dst prompts): canonical in 5/6 ordered pairs; $W \setminus W$ Jaccard 0.45–0.70, $C \setminus C$ 0.43–0.60 (partial overlap consistent with the task-conditional populations of §4.5), and zero role flips at any scale (full table in App. B.1). *Specificity*: 19/27 (70%) cancellers are rule-specific (rule-NLL drop $\geq 5 \times$ random-NLL drop), trending up with model size (410M 5/9, 1B 6/10, 1.4B 8/8; Cochran–Armitage $p = 0.049$); no above-chance overlap with induction top-10. Within-layer OV anti-alignment holds in 4/6 cells; both failures (hier-1B, mod-1.4B) occur on $|C| = 4$ pools, but $|C| = 4$ alone is not sufficient (hier-1.4B passes with the same canceller count; App. B.4). Edge-level comparison confirms real (W, C) pairs are significantly stronger than random layer-respecting pairs in 6/6 cells (Mann–Whitney $\text{BH-}q \leq 1.4 \times 10^{-11}$; App. B.4).

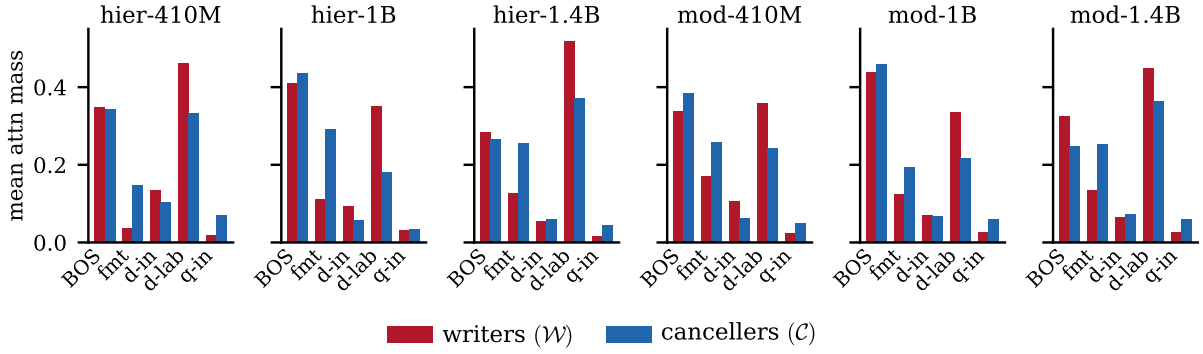


Figure 2: **Cancellers re-route attention from labels to format.** Mean per-head attention mass at the readout token for writers (red) vs cancellers (blue), per bucket, one panel per cell. Writers concentrate on demonstration labels; cancellers shift mass to format-prefix tokens — the direction of difference is consistent in 6/6 cells (per-cell mass in App. C.3, Table 11).

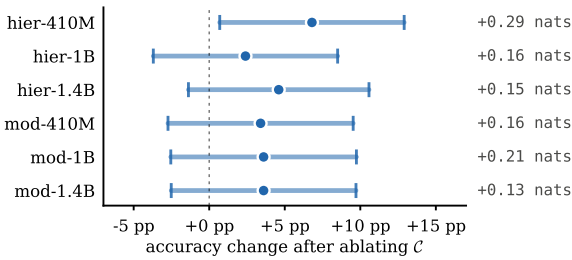


Figure 3: **Canceller ablation: actionable consequence.** Per-cell accuracy delta after zero-ablating $\mathcal{C}$, with paired-bootstrap 95% CI; logit-shift magnitude annotated on the right. Sign is positive in 6/6 main cells; CI excludes 0 on hier-410M. App. D.3 Table 21.

Layer geometry. Cancellers cluster in early-to-mid layers and 23/27 have an upstream writer in their own cell, ruling out late-stage suppression. Maximum writer layer reaches $L = 21$ – 23 in 24-layer Pythia; canceller layers stop at $L = 19$. Per-cell layout in App. B.6 (Figure 4).

4.5 Cross-template transfer to vocabulary ICL

Pythia-410M W/C labels transferred to two vocabulary-ICL templates (antonym, country-capital) preserve the canonical pattern in 3/4 pairs (App. C.5). The 1/4 hier→antonym “failure” flips cancellers to super-writers (-1.56 nats), exactly the pattern copy-suppression (McDougall et al., 2023) predicts when the demonstrated label *is* the answer: the W/C role is task-conditional, not intrinsic.

4.6 Scale and cross-architecture spot checks

The verdict extends to the rest of the Pythia ladder (2.8B/6.9B/12B) and three cross-architecture spot checks (Qwen2.5- $\{1.5, 7\}$ B, GPT-2-medium; modular only, App. B.5, Table 7). All 9 extension cells pass the signed-direction condition; 6.9B mod

and 12B hier shift to PARTIAL (one CI grazes 0). Combined with the main cells: 13/15 canonical, 2/15 partial, 0/15 sign-flipped.

4.7 Population-level mechanism rule-outs

Three named circuits are ruled out as universal explanations (App. C.6). *Rank-1 copy-suppression* (McDougall et al., 2023): 0/27 cancellers exceed the top-1 OV Frobenius threshold 0.5 (max 7.5%). *V-cascade* (Wang et al., 2023): ruled out in 4/6 cells (2/6 with an active counterfactual on the dominant upstream writer, 2/6 vacuous due to no upstream writer at $L < L_C$); weak-partial residual cascade in the remaining 2/6. *V-shuffle*: 86–113% collapse in 6/6, marking the mechanism as content-driven. W and C mean OV directions are nearly orthogonal (App. D.4, Table 22).

4.8 Mechanistic case study of L11.H4

The per-source decomposition of §4.3 identifies L11.H4 as the dominant canceller in both Pythia-410M rule cells ($\sim 100\%$ of its negative DLA from demonstration-label tokens). Three interventions narrow its mechanism (numerics in App. C.7). *V-shuffle* permutes the head’s value vectors across source positions and collapses cancellation by $\sim 82\%$ (hier) / $\sim 53\%$ (mod), so the head reads source content. The OV singular spectrum of $W_O W_V$ has top-1 Frobenius share 2.8% (top-10 25%, no rank-1 plateau): the read is distributed across many directions, ruling out rank-1 copy-suppression (McDougall et al., 2023). *V-composition* ablates the dominant upstream writer L10.H9 and leaves L11.H4’s DLA unchanged, ruling out S-inhibition V-cascade (Wang et al., 2023). *Cross-template sign-flip*: L11.H4’s solo ablation effect changes sign across templates (canceller on hier and mod, writer

Table 3: **Group lesion (zero strategy)**. Writer ablation shifts the readout -0.25 to -0.99 nats away from the correct label; canceller ablation $+0.13$ to $+0.29$ nats toward it; joint attenuates the writer effect by mean 48% (median 41%). Four-condition canonical (§3.3) holds 6/6 zero-strategy, 11/12 mean-strategy (mod-1.4B mean canceller below 0.10; App. B.2). $|\mathcal{F}|$ is the FV candidate set; only $\mathcal{W} \cup \mathcal{C}$ are ablated (weak heads with $|\text{direct\%}| < 5\%$ excluded).

Task	Model	$ \mathcal{F} $	$ \mathcal{W} $	$ \mathcal{C} $	$\Delta\ell_{\mathcal{W}}$	$\Delta\ell_{\mathcal{C}}$	$\Delta\ell_{\text{both}}$	verdict
hier	410M	22	8	6	-0.25	$+0.29$	$+0.00$	canonical
hier	1B	23	9	4	-0.56	$+0.16$	-0.43	canonical
hier	1.4B	19	10	4	-0.32	$+0.15$	-0.13	canonical
mod	410M	12	8	3	-0.39	$+0.16$	-0.14	canonical
mod	1B	16	8	6	-0.99	$+0.21$	-0.78	canonical
mod	1.4B	19	11	4	-0.49	$+0.13$	-0.38	canonical

on antonym, null on country-capital), so the role is task-conditional and the mechanism is consistent with “suppress what was demonstrated” rather than with unconditional logit suppression. L11.H4 is a *distributed-bias content reader*, distinct from both standard templates; we do not extrapolate this to the full canceller population.

5 Discussion

The FV literature treats high-magnitude heads as one functional class. We split the population into opposing sub-circuits at the same readout: sign-shuffle rejects homogeneity in 5/6 cells, QK reads differ in 6/6, OV directions are nearly orthogonal, 23/27 cancellers sit downstream of an upstream writer, and L11.H4 flips canceller→writer on antonym as “suppress what was demonstrated” predicts.

Connection to prior circuits and FV identification. IOI’s negative name-movers (Wang et al., 2023) and GPT-2 L10.H7’s copy-suppression (McDougall et al., 2023) are single-task circuits; we lift the phenomenon to a structural property of the FV head population (rank-1 holds for 0/27 cancellers; V-cascade ruled out in 4/6). Todd’s magnitude-only top- K is task-conditionally biased toward the locally-dominant sub-population (hier C 64% vs. W 4%; mod W 59% vs. C 8%), so any aggregation built on this ranking inherits a mixture of the two opposing roles.

Ablation, transplant, and additive steering are different operations. The W/C identification has different downstream consequences depending on the intervention. *Ablation* (zero out canceller outputs during forward pass) raises correct logit $+0.13$ – 0.29 nats and accuracy directionally $+2$ – 7 pp in 6/6 main cells (Figure 3; App. D.3);

this is the cleanest actionable consequence. *Transplant* (replace head outputs with their task-means) goes the opposite direction: writer-only transplant underperforms FV-mean transplant in accuracy in 6/6 cells (App. D.2), because the FV-mean transplant incidentally regularises cancellers’ per-prompt outputs while writer-only leaves cancellers active. *Additive steering* (inject a mean vector at the readout token) produces a larger absolute held-out logit shift for $v_{\mathcal{W}}$ than for $v_{\mathcal{FV}}$ in 6/6 main cells (App. D.1), but with α tuned on logit shift only; we make no positive claim on accuracy because under norm-matched injection with held-out α , the partition effect on accuracy is not established (see Limitations). The lesson: population-level identification tells us *which heads to intervene on*; the per-prompt nature of canceller behaviour means an aggregated steering vector is the wrong tool to clean it up under any α -free criterion (transplant: writer-only loses to FV-mean in 6/6 cells, App. D.2).

W/C composition predicts a transplant asymmetry. The canceller fraction in Todd’s top- K predicts the writer-only-vs-FV-mean transplant gap: hierarchical (64% cancellers in Todd top-20) shows a 6–12 pp accuracy gap; modular (8%) shows ≤ 2 pp (App. D.2). Cancellers’ mean activation is writer-orthogonal in OV space (App. D.4), so FV-mean transplant regularises their per-prompt outputs while writer-only leaves them active. **Conclusion.** The W/C split is structural, not prescriptive; it is invisible to magnitude-only ranking and inherited by any downstream FV-based aggregation (attribution graphs, ACDC, signed-DLA).

Limitations

Task scope. The primary analysis uses two synthetic ICL rule-following families (hierarchical, modular) with shared 4-shot surface form. Cross-

template transfer extends to two vocabulary-ICL templates (antonym, country-capital). All tasks are short, closed-label, and in-context-learnable from ≤ 5 demonstrations; we do not test long-form generation, open-ended NL classification at scale, or instruction-following.

Model coverage. Six Pythia main cells (410M, 1B, 1.4B $\times$ hier, mod) carry the full statistical pipeline. Extension cells (Pythia-2.8B/6.9B/12B; Qwen2.5- $\{1.5, 7\}$ B; GPT-2-medium) use a lighter protocol with $n_{\text{eval}} = 200$. We do not test Meta’s Llama series or instruction-tuned variants (Qwen2.5 shares the Llama-style attention architecture but is a separate model family). One main cell (mod-1.4B) is the boundary case on sign-shuffle ($p_{\text{emp}} = 0.104$); two extension cells (6.9B mod, 12B hier) shift to PARTIAL (one CI grazes 0).

Steering claims are limited. We make no positive claim that v_W improves practical steering. The only α -free comparison (transplant, App. D.2) shows writer-only is at best tied with FV-mean on accuracy, with the gap predicted by Todd’s top- K canceller fraction. Held-out α on the 6 main cells (App. D.1) is on logit shift only; accuracy under held-out α with norm-matched injection and path-patching W/C across all 15 cells is left to future work.

Mechanistic case study has narrow generality. L11.H4 (Pythia-410M) is the dominant canceller in both rule cells, but its “distributed-bias content reader” characterisation is established only at this one head. We do not claim the case-study mechanism generalises to the full 27 canceller (cell, head) pairs; we do show population-wide that rank-1 copy-suppression holds for 0/27 pairs, V-cascade is ruled out in 4/6 cells (2/6 active, 2/6 vacuous), and per-source DLA is content-dominated for 20/27 (74%) cancellers.

Statistical scope. Cross-cell aggregation uses Holm-Bonferroni FWER at $\alpha = 0.05$ over the 6 Pythia main cells (5/6 reject). The verdict matrix (six independent tests, App. B.7) reports per-test BH-FDR within each test family but not family-wise across the matrix as a whole, so the matrix is presented as a per-test pass-rate summary rather than a single composite test.

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A Methods and reproducibility

A.1 Statistical inventory

Every test the paper relies on. Bootstraps are paired-prompt percentile bootstraps with $B = 10,000$ unless noted; BH-FDR at $q = 0.10$ on raw p -values; display p -values floored at 1×10^{-6} .

A.2 Reproducibility

Models. Pythia checkpoints
EleutherAI/pythia-410m-deduped,
pythia-1b-deduped,
pythia-1.4b-deduped; final training step,
fp32, attn_implementation=eager.

Prompts. 4-shot template
 $f_0, f_1, Y \dots f_{0q}, f_{1q}$, with $f_0, f_1 \sim \text{Unif}\{0, \dots, 7\}$, $z \sim \text{Unif}\{0, 1, 2, 3\}$; format-prefix is the literal ", ". Per cell: 192 DLA prompts (seed 42), 200 PP prompts (seed 43), 500 eval prompts (seed 44). Logit aggregation: $\log \sum p$ over leading-space variants $\{ 'A', 'A' \}$ vs. $\{ 'B', 'B' \}$.

FV head set. Sign-preserving refined-DLA passes at $q = 0.10$ unioned with top- K by $|\Delta\ell|$, $K = \text{clip}(2n_{\text{FDR}}, 8, 50)$, intersected with PP gate $|\text{direct}\%| \geq 5\%$ to yield $\mathcal{F}$.

Path patching. Direct-path receiver-edge ablation of Wang et al. (2023); filter $|\Delta\ell^{\text{full}}| < 0.5$.

QK bucketing. Five deterministic bucket spans: BOS = $\{0\}$; format prefix (whitespace/punctuation between demos); demonstration input ((f_0, f_1) spans); demonstration label (Y tokens); query input ((f_{0q}, f_{1q})).

Edge-level null. 50 random head pairs (L_a, L_b) with $L_a < L_b$ from the full head population per cell.

Statistics. Bootstrap $B = 10,000$; sign-shuffle $n_{\text{perm}} = 10,000$; refined-DLA 20-seed permutation null; specificity $B = 5,000$. Thresholds ($\pm 5\%$ direct effect, 0.10 nat floor, $5\times$ rule/random ratio) fixed pre-analysis.

A.3 Threshold sensitivity

Headline 11/12 canonical is robust to $\pm 50\%$ perturbation of any single threshold ($\pm 5\%$ direct-effect gate, 0.10-nat magnitude floor, $5\times$ rule/random ratio); the single boundary case (mod-1.4B mean, canceller +0.09) flips to canonical at floors ≤ 0.09 .

Table 4: Statistical inventory.

Test	Null	Statistic	p -floor	Multcomp
Refined-DLA	20 label permutations	per-head signed mean	0.05	BH $q=0.10$
Path patching	$B=10,000$ paired bootstrap	ratio-of-means / two-sided	$1/B$	BH $q=0.10$
Group lesion	$B=10,000$ paired bootstrap	4-condition CI gate	$1/B$	—
Sign-shuffle null	$n_{\text{perm}}=10,000$ in-pool sign	$ W-C $ contrast	10^{-4}	BH $q=0.10$
Within-layer OV null	≥ 100 same-layer non-($W \cup C$) pairs	cosine $\leq$ obs	0.01	—
Edge-null ACDC	50 random layer-respecting pairs	MWU on $ \Delta\ell $	1×10^{-6}	BH $q=0.10$
Specificity ratio	$B=5,000$ canceller bootstrap	rule/random NLL ratio	—	—
Induction TOST	hypergeometric E , margin $\sqrt{\text{Var}_{\text{hg}}}$	two one-sided test	—	—
Cross-task verdict	sign+magnitude on dst eval (relaxed gate, §3.5)	4-condition variant	—	—
Split-half	$n_{\text{split}}=5$ random 50/50 splits	verdict-stable in $\geq n/2$ splits	—	—
Per-source DLA	per-token-source bucket attribution	$ \text{sink} $ vs. $ \text{content} $ DLA share	—	—
Vocab transfer	$B=2,000$ paired bootstrap, $n=100$	$\Delta\ell^{\text{correct}}$ shift / sign agreement	$1/B$	—
Cochran–Armitage	rule-specific rate by ordinal scale	trend z , two-sided	—	—

B Discovery and validation

B.1 Refined DLA and path patching: per-cell tables

Full per-head refined-DLA and PP tables released as JSONs; cell sizes in Table 3. End-to-end sum-to- $\Delta\ell$ reconstruction passes (rel. tol. 5%) in 9/9 tests for 6/6 cells.

B.2 Group lesion: zero and mean strategies

Table 5 mirrors Table 3 under mean ablation: 5/6 canonical, mod-1.4B writer-dominant (canceller +0.09 just below the 0.10 threshold).

FV-set sign-shuffle null per-cell. Table 6: observed $|W-C|$ contrast vs. in-pool sign-shuffle null with Gaussian back-stop and BH- q . 5/6 cells reach $q_{\text{emp}} \leq 0.014$; mod-1.4B is the boundary ($p_{\text{emp}}=0.104$, $p_{\text{Gauss}}=0.088$).

B.3 Asymmetric budgets and per-task breakdown

Per-head $|\text{direct}\%|$: $W = 20.3\%$ ($n = 54$), $C = 11.2\%$ ($n = 27$); per-head canceller shift is more uniform than writer (CV 0.19 vs. 0.56; canceller budget ~ 0.04 nats/head, task-independent). Modular elicits $\sim 1.7\times$ larger writer magnitudes than hierarchical ($|\Delta\bar{\ell}_{\text{W}}|=0.62$ vs. 0.38); cancellers not correspondingly amplified (0.17 vs. 0.20), so joint attenuation is lower on modular (36% vs. 61%).

B.4 ACDC edge ablation: pair-by-pair

Real-vs-null per cell: Mann–Whitney BH-FDR pass 6/6 at $q \leq 1.4 \times 10^{-11}$, rank-biserial $r \in [0.60, 0.97]$. Fisher canonical-rate BH-FDR pass: 1/6 (mod-410M, $q = 0.0016$; others raw $p \in [0.10, 0.21]$). Magnitude dominance is uni-

form; canonical-edge enrichment is task- and size-specific.

B.5 Scale and cross-architecture spot checks

B.6 Layer geometry of writers and cancellers

B.7 Per-cell verdict matrix

FWER across the verdict matrix. Holm–Bonferroni on sign-shuffle p -values $\{0.001, 0.001, 0.001, 0.014, 0.014, 0.104\}$ at family-wise $\alpha = 0.05$: 5/6 pass; mod-1.4B fails the final threshold. BH-FDR $q = 0.10$ yields the same conclusion. Other matrix entries are conjunctive threshold tests at fixed gates and are reported descriptively.

C Mechanism rule-outs

C.1 Specificity controls: per-cell tables

Table 9: per-cell rule-specific / indeterminate / generally-important breakdown for 27 cancellers and the top-10 induction overlap. Rule-specific rate sharpens with scale; no above-chance induction overlap. TOST equivalence margin is $\sqrt{\text{Var}_{\text{hg}}} \approx 0.5\text{--}0.7$ heads, so the test has limited power for 1–2 head intersections; the induction-distinction is also supported by QK-source attribution (§4.2).

Highest-specificity cancellers. L11.H4 (115 $\times$ hier-410M, 151 $\times$ mod-410M), L13.H1 (109 $\times$ hier-410M), L11.H2 (82 $\times$ mod-1.4B).

C.2 FV-overlap controls

Intersection of $\mathcal{F}$ with Todd et al. (2024)’s mean-ablation top- K at $K \in \{5, 10, 15, 20\}$ in Table 10. Total Jaccard 0.06–0.25 at top-20. The W vs. C asymmetry is task-dependent: hierarchical concentrates $\mathcal{C}$ in Todd’s top-20 (9/14 vs. writer 1/27);

Table 5: **Group lesion (mean strategy)**. 11/12 canonical across both strategies; the only miss is mod-1.4B mean.

Task	Model	$\Delta\ell_{\mathcal{W}}$	$\Delta\ell_{\mathcal{C}}$	$\Delta\ell_{\text{both}}$	verdict
hier	410M	-0.34	+0.29	-0.17	canonical
hier	1B	-0.44	+0.28	-0.34	canonical
hier	1.4B	-0.37	+0.22	-0.24	canonical
mod	410M	-0.31	+0.13	-0.23	canonical
mod	1B	-0.40	+0.26	-0.25	canonical
mod	1.4B	-0.35	+0.09	-0.26	writer-dominant

Table 6: **FV-set sign-shuffle null with Gaussian back-stop** ($n_{\text{perm}} = 10,000$). Empirical and Gaussian p -values; BH-corrected q -values across the 6-cell family at $q = 0.10$. Empirical p floored at $1/n_{\text{perm}} = 10^{-4}$.

Cell	observed	null p_{50}	null p_{95}	z	emp p	Gauss p	q_{emp}	q_{Gauss}
hier-410M	-0.435	-0.01	+0.24	-2.79	0.0024	5.2×10^{-3}	0.007	0.016
hier-1B	-0.607	-0.10	+0.24	-2.47	0.0047	1.3×10^{-2}	0.007	0.027
hier-1.4B	-0.275	-0.04	+0.13	-2.32	0.0035	2.0×10^{-2}	0.007	0.031
mod-410M	-0.361	-0.04	+0.17	-2.08	0.0116	3.7×10^{-2}	0.014	0.045
mod-1B	-0.683	-0.03	+0.35	-2.87	0.0010	4.1×10^{-3}	0.006	0.016
mod-1.4B	-0.223	+0.01	+0.22	-1.71	0.1040	8.8×10^{-2}	0.104	0.088

modular reverses (16/27 writers vs. 1/13 cancellers).

C.3 QK source distribution: per-cell attention mass

Per-cell mean attention mass by group and source bucket in Table 11. In 6/6 cells, demo-label mass drops from W to C (mean -0.13) and format-prefix rises (mean +0.13); other buckets shift by ≤ 0.05 .

C.4 Per-source DLA decomposition: cancellers vs. sinks

Per-canceller DLA attributed to source buckets via $\sum_{\text{pos}} \alpha_{\text{pos}} W_O W_V r_{\text{pos}}$ projected onto $u(x)$. Per-cell summary in Table 12; full JSON released.

Named-head detail. L11.H4: $\sim 100\%$ demo-label sourced in both cells.

Comparison to GPT-2 copy-suppression (McDougall et al., 2023). Table 13.

C.5 Cross-template transfer to vocabulary ICL

Two vocab-ICL tasks (antonym, 60 pairs; country-capital, 50 pairs; $k = 4$, format word: answer). Zero-strategy ablation, $n_{\text{eval}} = 100$, $B = 2,000$. Sub-population in Table 14; L11.H4 solo in Table 15.

C.6 Multi-cell mechanism rule-outs (rank-1 / V-cascade / V-shuffle)

Three named-mechanism rule-outs extended to the dominant canceller per cell: rank-1 (top-1

OV Frobenius share, all cancellers); V-cascade (zero-ablate top-3 anti-aligned upstream writers, $n_{\text{boot}} = 2000$ BCa); V-shuffle (relative collapse $\geq 0.5 = \text{content-driven}$).

Rank-1 ruled out population-wide (0/27 cancellers exceed 7.5% top-1 OV vs. 50% threshold). V-shuffle content-driven in 6/6 (rel-collapse 0.86–1.13). V-cascade ruled out 4/6; hier-1B and mod-1B show a small residual cascade ($\Delta \text{DLA} \in [+0.02, +0.03]$) from a near-orthogonal upstream writer (cosine $[-0.04, -0.01]$).

C.7 Mechanistic case study: V-shuffle, OV spectrum, V-composition

Three single-head experiments on Pythia-410M L11.H4.

D W/C aggregation: clean comparisons

We report two methodologically clean comparisons of v_{FV} vs. $v_{\mathcal{W}}$. The (in-sample α , per-vector tuning, top- K heuristic W/C on extension cells) variant from an earlier pilot is excluded as confounded; results below use the path-patching W/C partition and either held-out α or α -free transplant.

D.1 Held-out α on 6 main cells

192/500 calibration-evaluation split; per-cell α selected on calibration over the grid $\{0.25, 0.5, 1, 2, 4, 8, 16\}$ to maximise $\overline{\Delta\ell}$, evaluated on disjoint hold-out. All three vectors injected at layer $\lfloor n_{\text{layers}}/2 \rfloor$ on the readout token.

Table 7: **Verdict across architectures and scale.** Group-lesion W / C / joint logit shifts (mean strategy, $n_{\text{eval}}=200$). $W < 0$, $C > 0$ in all 9 extension cells; combined with the 6 main cells: 13/15 canonical, 2/15 partial (signed direction holds, one CI grazes 0).

Family / Arch	Size	Task	W_{shift}	C_{shift}	joint	verdict
Pythia (NeoX)	2.8B	hier	-0.30	+0.28	+0.01	canonical
Pythia	2.8B	mod	-0.20	+0.19	+0.01	canonical
Pythia	6.9B	hier	-0.15	+0.13	+0.02	canonical
Pythia	6.9B	mod	-0.11	+0.05	-0.01	partial
Pythia	12B	hier	-0.10	+0.17	-0.01	partial
Pythia	12B	mod	-0.11	+0.18	+0.11	canonical
Qwen2.5 (Llama-style)	1.5B	mod	-0.44	+0.11	-0.03	canonical
Qwen2.5	7B	mod	-0.11	+0.12	+0.10	canonical
GPT-2-medium (Conv1D)	355M	mod	-0.33	+0.38	+0.03	canonical

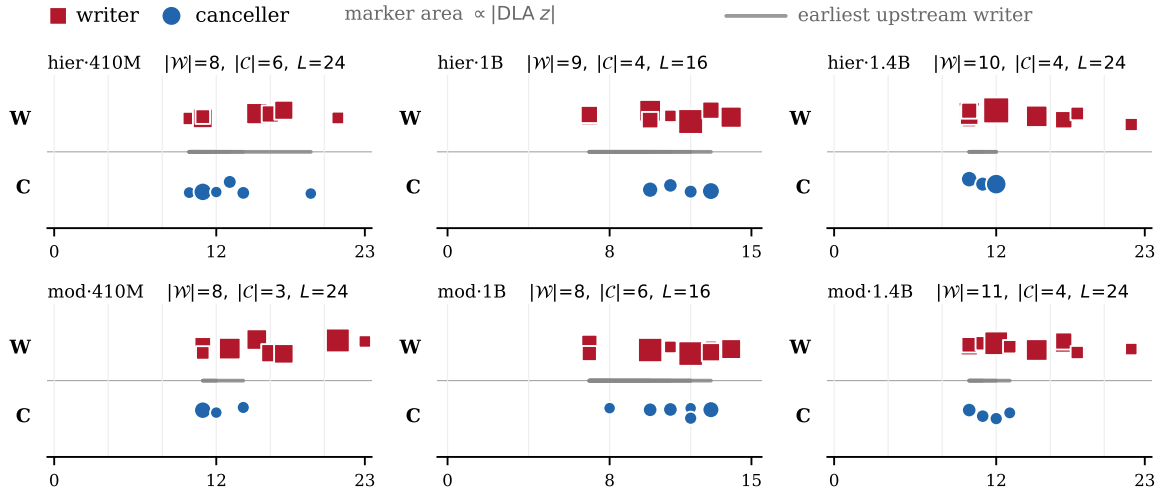


Figure 4: **Layer distribution of writers and cancellers.** Marker area $\propto z$ -scored $|DLA|$; gray lines connect each canceller to its earliest upstream writer. Cancellers cluster in early-to-mid layers and 23/27 have an upstream writer, ruling out late-stage suppression.

D.2 Transplant accuracy: writer-only vs FV-mean

Each head’s activation slot is *replaced* with its task-mean ($n_{\text{eval}} = 200$). The asymmetry $\Delta\text{acc}(W) - \Delta\text{acc}(F)$ is ≤ 0 in 6/6 cells, the opposite direction from the additive-injection logit-shift result. Replacement neutralises cancellers’ per-prompt outputs; sign-aware drop does not.

D.3 Per-cell ablation accuracy effects

Group-lesion accuracy effects on the 6 main cells. Sign of the shift matches the W/C direct-effect sign in 6/6 cells; $\Delta\text{acc}(W)$ CI excludes 0 in 3/6, $\Delta\text{acc}(C)$ CI excludes 0 in 1/6.

D.4 Cancellers are writer-orthogonal in OV space

Decomposition $\bar{a}_C = \alpha_{\parallel} \bar{a}_W + \bar{a}_C^{\perp}$, perpendicular fraction 0.84–1.00 (mean 0.96, 95% CI [0.91, 0.99]); $\cos$ mean -0.18, 95% CI

$[-0.36, -0.03]$.

E Robustness controls

E.1 Head-randomised control: the tautology rule-out

Tests whether the canonical 4-condition verdict is a tautology of sign-partitioning a high-magnitude population. Per cell: $n_{\text{eval}} = 100$, top- $K = 14$ FV by $|DLA|$ vs. 10 random seeds of $|DLA|$ -matched non-FV partitions, zero-strategy group lesions on both. Table 23 reports per-cell aggregates.

FV ablations are 5.5 $\times$ (W) and 5.2 $\times$ (C) larger than $|DLA|$ -matched random, attenuation gap 2.5 $\times$ (0.047 vs. 0.019 nats). The sign part is mechanically forced; the loose attenuation flag fires for 77% of random seeds, so magnitude (not sign) distinguishes FV from non-FV. Matching is rank-closest (non-FV $|DLA|$ averages 0.04 vs. FV 0.20 nats); the mod-1.4B FV/random gap < 1 is consistent with its weak canceller pool ($|C| = 4$).

Table 8: **Per-cell verdict matrix across six independent tests.** ✓ pass, ✗ fail, ○ boundary. The cancellation signature holds in 6/6 cells under the strongest test (zero-strategy 4-condition); secondary tests rule out role-flipping, induction overlap, and writer/canceller equivalence on 4–6 of 6 cells.

Cell	Zero 4c	Mean 4c	Sign-shuf q	Split-half	Cross-task 4c	OV gate	Fisher edge q
hier-410M	✓	✓	✓	✓	✓	✓	✗
hier-1B	✓	✓	✓	✓	○ W-dom	✗	✗
hier-1.4B	✓	✓	✓	✓	✓	✓	✗
mod-410M	✓	✓	✓	✓	✓	✓	✓
mod-1B	✓	✓	✓	✓	✓	✓	✗
mod-1.4B	✓	○ W-dom	○ ($q=0.104$)	✓	✓	✗	✗
pass	6/6	5/6	5/6	6/6	5/6	4/6	1/6

Table 9: **Per-cell specificity and induction-overlap controls.** *rule-spec*: cancellers with rule/random NLL ratio ≥ 5 . *ind. ovlp.*: $|\mathcal{F} \cap \text{top-10}_{\text{ind}}|/|\mathcal{F}|$ (intersection of FV head set with top-10 induction heads).

Cell	$ \mathcal{C} $	rule-spec	indet	gen-imp	ind. ovlp.
hier-410M	6	4	0	2	1/22
hier-1B	4	3	1	0	3/23
hier-1.4B	4	4	0	0	0/19
mod-410M	3	1	1	1	0/12
mod-1B	6	3	2	1	1/16
mod-1.4B	4	4	0	0	0/19
total	27	19 (70%)	4	4	—

Table 10: **Full 6-cell FV-overlap.** W/C set from the PP-validated set used throughout the paper; FV top- K from Todd’s mean-ablation identification on the same task. The W- vs C-asymmetry column shows W-in-FV / C-in-FV at $K=20$; (C-heavy) marks cells where cancellers concentrate in Todd’s top-20, (W-heavy) marks the opposite.

Cell	$ \mathcal{W} $	$ \mathcal{C} $	Jaccard	W in FV-20	C in FV-20	pattern
hier-410M	8	6	0.06	0	2	C-heavy
hier-1B	9	4	0.18	1	4	C-heavy
hier-1.4B	10	4	0.10	0	3	C-heavy
mod-410M	8	3	0.19	5	0	W-heavy
mod-1B	8	6	0.17	5	0	W-heavy
mod-1.4B	11	4	0.25	6	1	W-heavy
hier total	27	14	—	1 (4%)	9 (64%)	—
mod total	27	13	—	16 (59%)	1 (8%)	—

Table 11: **Per-cell mean attention mass at the final query token, by W/C group and source bucket.** Each row is one (group, cell). n is the number of heads in the group. demo-lab/format-prefix differences are the load-bearing buckets of Figure 2.

Cell	Grp	n	BOS	format	d-in	d-lab	q-in
hier-410M	W	8	0.35	0.04	0.14	0.46	0.02
	C	6	0.34	0.15	0.11	0.33	0.07
hier-1B	W	9	0.41	0.11	0.09	0.35	0.03
	C	4	0.44	0.29	0.06	0.18	0.03
hier-1.4B	W	10	0.28	0.13	0.05	0.52	0.02
	C	4	0.27	0.26	0.06	0.37	0.05
mod-410M	W	8	0.34	0.17	0.11	0.36	0.02
	C	3	0.38	0.26	0.06	0.24	0.05
mod-1B	W	8	0.44	0.12	0.07	0.34	0.03
	C	6	0.46	0.20	0.07	0.22	0.06
mod-1.4B	W	11	0.32	0.13	0.06	0.45	0.03
	C	4	0.25	0.25	0.07	0.36	0.06

Table 12: **Per-cell content vs. sink decomposition of cancellers.** A canceller is content-driven if $|\text{BOS} + \text{fmt}| < |\text{demo}_{\text{in}} + \text{demo}_{\text{lab}} + \text{query}_{\text{in}}|$ in DLA contribution, sink-driven otherwise. $\sum \Delta\ell$ sums total per-head DLA over each sub-population. Content cancellers carry the headline magnitude in 5/6 cells; the hier-1B exception splits 2/4 heads as sink-classified. Stable cross-task sink heads: L14.H3 (both 410M), L10.H2 (both 1B), L11.H2 (both 1.4B).

Cell	content $ \mathcal{C} $	sink $ \mathcal{C} $	content $\sum \Delta\ell$	sink $\sum \Delta\ell$
hier-410M	5	1 (L14.H3)	-0.38	-0.03
hier-1B	2	2 (L10.H2, L12.H5)	-0.15	-0.13
hier-1.4B	3	1 (L11.H2)	-0.25	-0.04
mod-410M	2	1 (L14.H3)	-0.26	-0.02
mod-1B	5	1 (L10.H2)	-0.21	-0.02
mod-1.4B	3	1 (L11.H2)	-0.16	+0.00

Table 13: **L11.H4 (this work) vs. GPT-2 L10.H7 (McDougall et al., 2023).** Both are named, single-head, content-driven negative-direction circuits. We do not yet test whether L11.H4 exhibits L10.H7’s full copy-suppression signature on copy-task contexts; this is left as future work.

Property	GPT-2 L10.H7 (McDougall ’23)	Pythia-410M L11.H4 (this work)
Layer-relative	$L = 10/12$, late-mid	$L = 11/24$, early-mid
Direct effect	negative on attended-token logit	negative on correct rule-label logit
QK signature	attends-to copies of query token	invariant across rule tasks
OV map	rank-1 promote/suppress	cos -0.228 vs. writer L10.H9 (not yet rank-resolved)
Tasks tested	> 200 data slices	2 rule + 2 vocab ICL tasks (this work)
Sign flip on negation	not reported	flips canceller $\rightarrow$ writer on antonym
Copy-task behaviour	defining (copy-suppression)	not yet tested

Table 14: **Cross-template W/C transfer: 3/4 canonical pairs.** Mean shift in correct-token logit; positive = ablation helps correct; negative = ablation hurts. Canonical pattern requires (i) writers negative, (ii) cancellers positive, (iii) joint attenuates writers.

src cell	target task	writer $\Delta\ell$	canc $\Delta\ell$	joint $\Delta\ell$	verdict
hier-410M	antonym	-0.48	-1.56	-2.00	writer-dominant (fail)
hier-410M	country-capital	-0.52	+0.14	-0.37	canonical
mod-410M	antonym	-0.33	+0.03	-0.24	canonical (C-CI just over 0)
mod-410M	country-capital	-0.59	+0.23	-0.36	canonical

Table 15: **L11.H4 solo ablation on vocabulary tasks.** The *address* transfers but the *role* is task-conditional: on antonym L11.H4 acts as a writer (CI excludes 0 negative); on country-capital it has no significant effect.

target task	mean shift	95% CI	verdict
antonym	-0.069	$[-0.109, -0.028]$	writer (CI excludes 0)
country-capital	-0.005	$[-0.033, +0.025]$	null (CI includes 0)

Table 16: **Multi-cell mechanism rule-outs.** $|\mathcal{C}|$: cancellers in the cell. $\sigma_1^{\max}/\|\text{OV}\|_F^2$: max top-1 OV Frobenius share across all cancellers in the cell. $n_{\text{rank-1}}$: number of cancellers with top-1 OV ≥ 0.5 . $\Delta\ell$ V-shuffle: V-shuffle relative collapse on the top canceller. V-cascade verdict on the top canceller from ablating the top-3 anti-aligned upstream writers (cos < 0): RULED OUT means CIs include 0 for all 3; PARTIAL means ≥ 1 writer’s CI excludes 0; VACUOUS means no upstream writer at $L < L_{\text{canceller}}$.

Cell	top canc.	$ \mathcal{C} $	$\sigma_1^{\max}$	$n_{\text{rank-1}}$	$\Delta\ell$ V-shuf rel	V-cascade
hier-410M	L11.H4	6	0.070	0/6	0.86	RULED OUT
hier-1B	L13.H1	4	0.075	0/4	1.00	PARTIAL (L7.H2 cos = -0.04)
hier-1.4B	L12.H7	4	0.030	0/4	0.98	RULED OUT
mod-410M	L11.H4	3	0.070	0/3	1.13	VACUOUS (no upstream W)
mod-1B	L13.H1	6	0.075	0/6	1.05	PARTIAL (L7.H0 cos = -0.01)
mod-1.4B	L10.H7	4	0.030	0/4	0.95	VACUOUS (no upstream W)
cells		27	≤ 0.075	0/27	≥ 0.86 in 6/6	4/6 ruled out, 2/6 partial

Table 17: **V-shuffle and V-composition diffs on L11.H4 DLA.** V-shuffle: random permutation of per-head V across source positions ($n_{\text{eval}} = 200$, 3 seeds, paired-prompt bootstrap). V-composition: zero-ablate L10.H9 (dominant upstream writer). V-shuffle CIs exclude zero in both cells (content-driven); V-composition CIs include zero (parallel, not cascaded).

Intervention	Cell	baseline DLA	post DLA	diff	95% CI
V-shuffle	hier-410M	-0.216	-0.040	+0.176	[+0.080, +0.279]
	mod-410M	-0.215	-0.103	+0.113	[+0.015, +0.206]
V-composition	hier-410M	-0.216	-0.209	+0.007	[-0.001, +0.015]
	mod-410M	-0.215	-0.211	+0.005	[-0.002, +0.011]

Table 18: **OV singular spectrum.** L11.H4’s $W_O W_V \in \mathbb{R}^{1024 \times 1024}$, $\|OV\|_F = 7.64$. The top-1 left-singular mode anti-aligns with task-mean $u(x)$ at $\cos = -0.13$, but its Frobenius share is only 2.8% (top-5: 13%; top-10: 25%); the map is dense, with no rank-1 plateau (McDougall et al., 2023).

mode k	σ_k	$\cos(u_k, u)$	signed contrib. $\sigma_k \cos$
0	1.27	-0.13	-0.16
1	1.26	+0.03	+0.04
2	1.24	+0.01	+0.01
3	1.23	+0.07	+0.09
4	1.21	-0.00	-0.00

Table 19: Held-out logit shift at calibrated α .

Task	Model	$\Delta\ell(v_{\text{FV}})$	$\Delta\ell(v_{\text{W}})$	$\Delta\ell(v_{\text{PCA}})$
hier	410M	-0.002	-0.003	-0.007
hier	1B	-0.002	-0.007	-0.003
hier	1.4B	-0.004	-0.022	+0.003
mod	410M	+0.095	+0.159	-0.002
mod	1B	-0.010	+0.012	-0.002
mod	1.4B	-0.005	+0.097	-0.032

Table 20: Transplant accuracy ($n_{\text{eval}} = 200$).

Cell	baseline	$\Delta\text{acc}(\mathcal{F})$	$\Delta\text{acc}(\mathcal{W})$	asymmetry
hier-410M	0.525	+0.025	-0.040	-0.065
hier-1B	0.595	+0.000	-0.100	-0.100
hier-1.4B	0.600	+0.005	-0.110	-0.115
mod-410M	0.480	+0.090	+0.085	-0.005
mod-1B	0.515	+0.065	+0.060	-0.005
mod-1.4B	0.520	+0.085	+0.065	-0.020

Table 21: Δacc with Wald 95% CI.

Cell	baseline	$\Delta\text{acc}(\mathcal{W})$	$\Delta\text{acc}(\mathcal{C})$	$\Delta\text{acc}(\mathcal{WUC})$
hier-410M	0.55	-0.05 [-0.11, +0.01]	+0.07 [+0.01, +0.13]	+0.03 [-0.03, +0.09]
hier-1B	0.58	-0.16 [-0.22, -0.10]	+0.02 [-0.04, +0.09]	-0.12 [-0.18, -0.06]
hier-1.4B	0.61	-0.07 [-0.13, -0.01]	+0.05 [-0.01, +0.11]	0.00 [-0.07, +0.06]
mod-410M	0.56	-0.01 [-0.07, +0.05]	+0.03 [-0.03, +0.10]	+0.08 [+0.02, +0.15]
mod-1B	0.55	-0.23 [-0.29, -0.17]	+0.04 [-0.03, +0.10]	-0.16 [-0.22, -0.10]
mod-1.4B	0.56	-0.01 [-0.07, +0.06]	+0.04 [-0.03, +0.10]	-0.06 [-0.12, +0.00]

Table 22: Writer/canceller OV geometry on the 6 main cells.

cell	$\cos(\mathcal{W}, \mathcal{C})$	perp. frac.	$\alpha_{\parallel}$
hier-410M	+0.05	1.00	+0.08
hier-1B	-0.32	0.95	-1.02
hier-1.4B	+0.03	1.00	+0.04
mod-410M	-0.54	0.84	-1.91
mod-1B	-0.24	0.97	-0.22
mod-1.4B	-0.09	1.00	-0.26

Table 23: **Head-randomised control across 6 Pythia cells.** For each cell we report the FV-set’s writer/canceller/joint ablation logit shifts and the mean over 10 random $|\text{DLA}|$ -matched non-FV partitions. Sign convention: writer = $\text{DLA} > 0$ (pushes correct $\uparrow$), canceller = $\text{DLA} < 0$ (pushes correct $\downarrow$); ablating writers shifts $-$, ablating cancellers shifts $+$. Magnitudes $|W|$, $|C|$, $\text{gap} = |W| + |C| - |\text{both}|$. Last column: fraction of random matched seeds whose joint ablation passes the loose binary attenuation criterion $|\text{both}| < |W| + |C| - 0.005$.

Cell	FV-set			Random matched (mean over 10 seeds)			atten frac	FV/rand gap
	$ W $	$ C $	gap	$ W $	$ C $	gap		
hier-410M	0.184	0.068	0.034	0.037	0.034	0.008	8/10	4.3×
hier-1B	0.161	0.206	0.101	0.039	0.019	0.019	6/10	5.3×
hier-1.4B	0.097	0.097	0.082	0.046	0.031	0.042	9/10	1.9×
mod-410M	0.166	0.349	0.013	0.019	0.007	0.005	6/10	2.6×
mod-1B	0.148	0.311	0.045	0.015	0.043	0.017	8/10	2.6×
mod-1.4B	0.102	0.172	0.010	0.060	0.031	0.023	9/10	0.4×
mean	0.143	0.200	0.047	0.036	0.027	0.019	46/60 (77%)	2.5×